

Roole at the SMT-COMP 2026

Jan Onderka
University of Freiburg
Freiburg, Germany
onderka@cs.uni-freiburg.de

Armin Biere
University of Freiburg
Freiburg, Germany
biere@cs.uni-freiburg.de

Abstract—We present **Roole**, a Satisfiability Modulo Theories (SMT) solver for the theory of quantifier-free bitvectors, as submitted to the SMT-COMP 2026.

I. INTRODUCTION

Roole [1] is a new SMT solver for the theory of quantifier-free bitvectors based on simplified Input-based Three-Valued Abstraction Refinement [2] combined with DPLL-style search with learning and backjumping. **Roole** computes bitvector operations in three-valued bitvector abstraction [3], using no bitblasting nor any underlying SAT solvers. It is dual-licensed under both the MIT and Apache-2.0 licence.

Roole is written in Rust and is designed with focus on certified solving, with the proof-checker **Roolean** [4] written in Lean that can check that the results are sound by essentially solving once again with the aid of *oracle certificates* from **Roole** that guide its choices for better performance, the soundness of **Roolean** results proven through Lean theorems. However, as SMT-COMP does not require results to be certified, we submit **Roole** as a standalone solver.

II. APPROACH

We based **Roole** on our previous work on Input-based Three-Valued Abstraction Refinement [3], using the fact that SMT solving of quantifier-free formulas is similar to model-checking the property $\text{EX}[p]$ (there exists a path where p holds in the next state), simplifying to the following.

Consider a function $f : V_1 \rightarrow V_2$. Introduce abstract values in sets $\hat{V}_1$ and $\hat{V}_2$ related by the *concretization function* $\gamma : \hat{V}_\square \rightarrow 2^{V_\square}$ that determines the concrete values overapproximated by the abstract value. The function $\hat{f} : \hat{V}_1 \rightarrow \hat{V}_2$ soundly overapproximates f exactly if

$$\forall \hat{v} \in \hat{V}_1. \forall v \in \gamma(\hat{v}). f(v) \in \gamma(\hat{f}(\hat{v})). \quad (1)$$

Once sound abstract functions are implemented for basic operations, they can be composed while retaining soundness. Finally, considering f as the term determining satisfaction of the formula by a given assignment $v \in V_1$, the result of checking satisfiability is equal to

$$\exists v \in V_1. f(v). \quad (2)$$

Brute-force computation using f can then be replaced by computation using its sound abstraction $\hat{f}$ on a set of abstract assignments $\hat{A} \subseteq \hat{V}_1$ provided that no spurious abstract values are considered ($\forall \hat{a} \in \hat{A}. \gamma(\hat{a}) \neq \emptyset$) and all concrete values are considered ($(\bigcup_{\hat{a} \in \hat{A}} \gamma(\hat{a})) = V_1$).

III. IMPLEMENTATION

The efficiency of the approach depends on balancing the computation speed of abstract operations and the number of abstract assignments needed to consider.

Currently, we use the three-valued bitvector domain with fast computation of arithmetic operations [3]. To search through the space of the abstract assignments for those where satisfiability or unsatisfiability of all overapproximated concrete assignments is shown, we use a search algorithm in the style of a Davis-Putnam-Logemann-Loveland (DPLL) procedure with additional basic machinery for backjumping and learning of abstract assignments found to be unsatisfiable, inspired by Conflict-Driven Clause Learning (CDCL).

We note that the current implementation is a first stepping stone, without the use of commonly used techniques such as preprocessing or sophisticated CDCL techniques, more powerful domains, or any fine-tuning.

REFERENCES

- [1] J. Onderka, “GitHub - Roole,” <https://github.com/onderjan/roole>, 2026, this paper discusses the version 0.1.1. [Accessed 2026-05-26].
- [2] J. Onderka and S. Ratschan, “Input-based three-valued abstraction refinement,” in *Verification, Model Checking, and Abstract Interpretation - 27th International Conference, VMCAI 2026, Rennes, France, January 12-13, 2026, Proceedings*, ser. Lecture Notes in Computer Science, Y. Chen, T. P. Jensen, and O. Lengál, Eds. Springer, 2026, pp. 237–262. [Online]. Available: https://doi.org/10.1007/978-3-032-15700-3_12
- [3] —, “Fast three-valued abstract bit-vector arithmetic,” in *Verification, Model Checking, and Abstract Interpretation - 23rd International Conference, VMCAI 2022, Philadelphia, PA, USA, January 16-18, 2022, Proceedings*, ser. Lecture Notes in Computer Science, B. Finkbeiner and T. Wies, Eds. Springer, 2022, pp. 242–262. [Online]. Available: https://doi.org/10.1007/978-3-030-94583-1_12
- [4] J. Onderka, “GitHub - Roolean,” <https://github.com/onderjan/roolean>, 2026, this paper discusses the version 0.1.0. [Accessed 2026-05-26].